# WaypointGen
# Learning Language-Grounded Waypoint Navigation

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## Motivation

- **Challenge:** Agricultural robot navigation requires extensive site-specific tuning. Traditional SLAM and perception degrade in unstructured vegetation and lack interfaces for natural language (NL) constraints.
- **WaypointGen:** A map-free, language-conditioned path planner operating directly in 2D BEV space, using VLM Reasoning to choose MPPI Trajectories that respect NL Navigation command constraints and preferences.
- **Pipeline:** Parses ROI constraints from NL commands via open-vocabulary grounding to synthesize constraint-aware waypoints. A novel semi-supervised labeling pipeline generates our dataset.
- **Impact:** Edge-deployable across diverse farm environments.

## Method: NL Command Generation

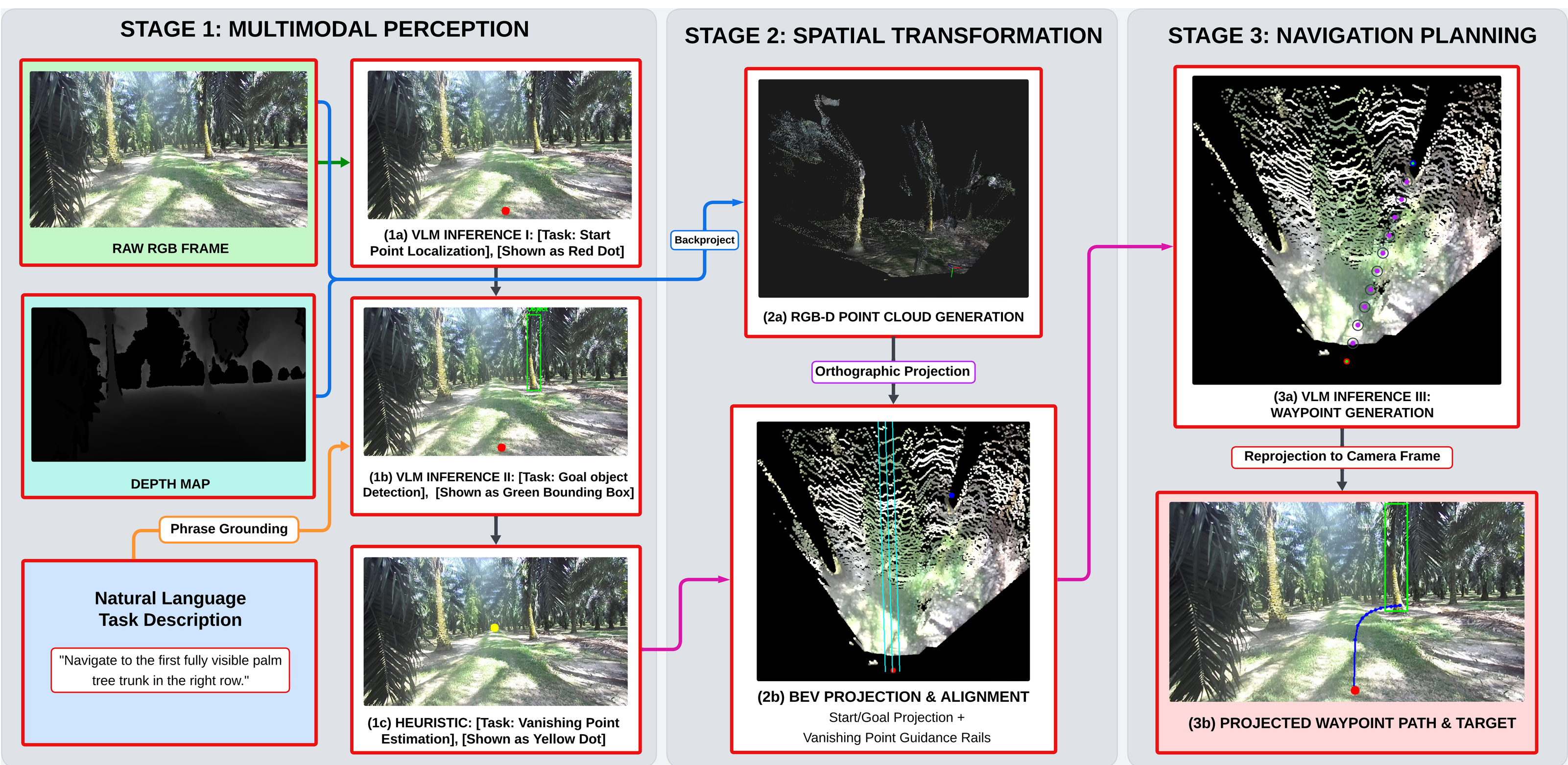


Fig. 1: **WaypointGen Inference** (1) DINO-X detects open-vocabulary objects; SAM3 produces instance masks and centroids; DAM generates mask-level NL descriptions. (2) Using a small human-labeled seed, Qwen3-MoE generates candidate navigation commands; Qwen3-235B filters them as a VLM judge. (3) For each command, Qwen3 selects a target ROI, MPPI proposes 7 feasible BEV trajectories, and Qwen3 selects the best one to export waypoint supervision.

**Goal:** Automatically generate grounded NL commands and 2D BEV waypoints.

### Step 1: Data Preprocessing:

- **Index:** Index of images and NL instructions.
- **Commands:** Image-specific NL commands mapping to objects (generated via QWEN-30B-A3B).
- **Templates:** Paths to Single Target ROI and NL Templates for CSL SC.
- **Perception+BEV:** SAM3 masks + DAM summaries, RGBD data, and BEV Image overlaid with start/goal points and camera viewpoint.

### Step 2: Human Annotation & In-Context Learning Setup:

- Human annotators mark 5 images each in the "Egocentric Human View" based on NL templates for in-context learning.
- MPPI trajectory overlays per-image are exported for datasets: Citrus, Palm, Solar, Corn, Soybean, and TreeScope.

## Method: Semi-supervised dataset generation

### Step 3: Semi-Supervised Dataset Generation Pipeline:

1. **Inference 1 (Target Grounding):** QWEN-MoE-30B-A3B selects a *single Target ROI* based on the input NL command and available bounding boxes/-masks.
2. **Inference 2 (Trajectory Selection):** QWEN-MoE-30B-A3B dynamically selects the best path from *7 candidate MPPI trajectories* pointing to the target ROI constraint.
3. **Export:** Waypoints generated to navigate to target ROI.

## Related Work

### Baselines

- **Zero shot VLMs**: Zero-shot VLM waypoint generation directly from raw images.
- **GeNIE [4]**: Generalizable VLM navigation. **Limitation:** Struggles to strictly adhere to fine-grained geometric and structural constraints necessary for agricultural rows. **WaypointGen** is better as it enforces constraints by explicitly verifying and selecting feasible MPPI trajectories in 2D BEV space prior to execution.
- **3D Reconstruction**: Dense point cloud fusion with RRT* planning.

### Datasets

1. **Palm plantation:** 500 m rows with 30+ trees and water channels.
2. **Solar farm:** Panel arrays, maintenance paths, and vegetation.
3. **Greenhouse (High tunnels):** Variable row spacing, Dynamic Lighting, Dynamically moving obstacles with Rigid Non Traversable occluded Plant Stems and Pliable Traversable leaves, uneven terrain.
4. **Corn and Soybean fields:** SLAM benchmark with stereo images, IMU, odometry, and GPS across growth stages [2].
5. **CitrusFarm:** Multimodal dataset featuring 9 sensing modalities across 7.5 km of trajectories in citrus tree farms [3].
6. **TreeScope:** LiDAR mapping dataset from forests and orchards for evaluating 3D mapping in natural vegetation [1].

## Conclusion and Future Work

- **Conclusion:** WaypointGen is a language-conditioned path planner that generates constraint-aware waypoints in 2D BEV space, avoiding dense 3D reconstruction.
- **Future Work:** Support Multiple Target ROIs, MPPI Trajectory selection for subpaths in long path navigation, and predict temporal waypoint orders for complex navigation.

## References

[1] Derek Cheng et al. *TreeScope: An Agricultural Robotics Dataset for LiDAR-Based Mapping of Trees in Forests and Orchards.* 2023. arXiv: 2310.02162 [cs.RO].
[2] Jose Cuaran et al. "Under-canopy dataset for advancing simultaneous localization and mapping in agricultural robotics". In: *The International Journal of Robotics Research* 43.6 (2024), pp. 739–749. DOI: 10.1177/02783649231215372.
[3] Hanzhe Teng et al. "Multimodal Dataset for Localization, Mapping and Crop Monitoring in Citrus Tree Farms". In: *International Symposium on Visual Computing*. 2023, pp. 571–582.
[4] Jiaming Wang et al. "GeNIE: A Generalizable Navigation System for In-the-Wild Environments". In: *IEEE Robotics and Automation Letters* 10.12 (2025), pp. 12628–12635. DOI: 10.1109/LRA.2025.3623038.

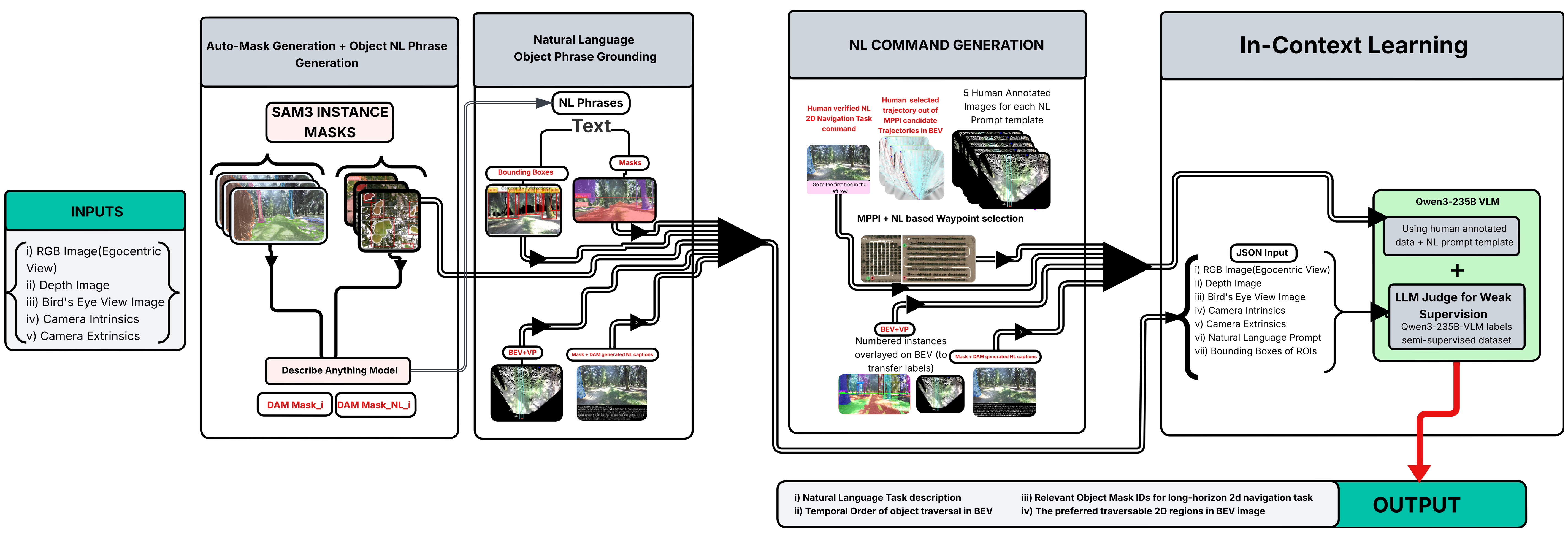


Fig. 2: Semi-supervised dataset generation pipeline overview. (i) SAM3 and DAM generate Masks and NL descriptions. (ii) QWEN3 is used to generate relevant 2D NL Navigation task descriptions. (iii) Human annotations are used for